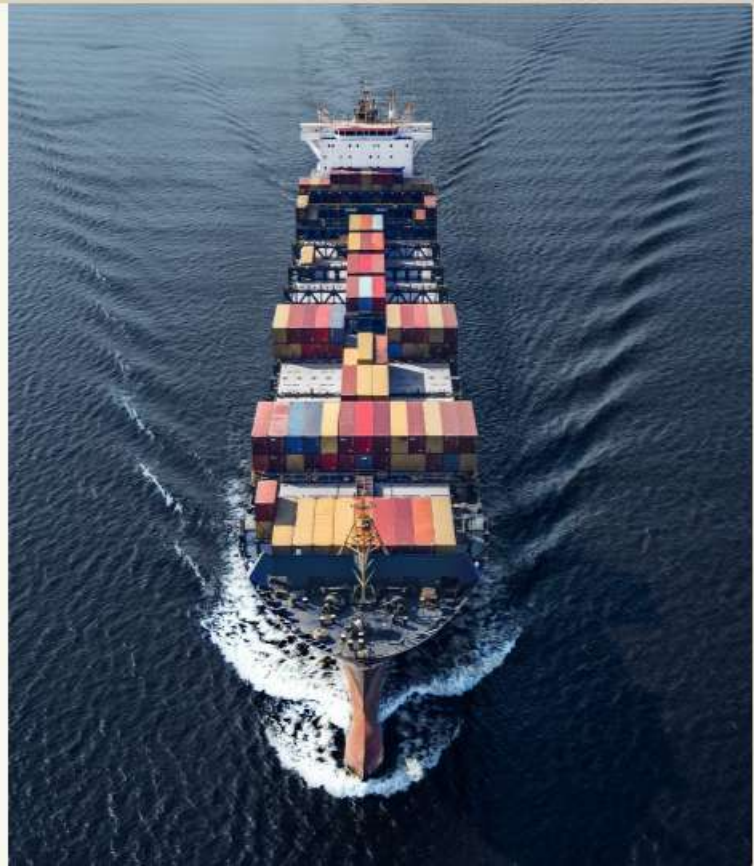


PortPulse

Port Congestion Prediction &
Operations Intelligence

Team Surfers



The Congestion Problem Is Costly — and Predictable

Port terminals operate under constant pressure: vessel queues grow, berths sit idle, and decisions arrive too late. The result is a cascade of disruption — delayed cargo, wasted capacity, and operational uncertainty.

Terminal Operators

Reactive berth assignments and unplanned vessel delays

Berth Planners

No visibility into congestion windows or wait-time risk

Shipping Teams

Unpredictable arrivals disrupt scheduling and cargo flow

Logistics Coordinators

Downstream delays with no early-warning signal

From Reactive Handling to Proactive Visibility

PortPulse is an platform that ingests vessel schedules and berth capacity data, then transforms it into actionable recommendations — giving planners the foresight to act before congestion forms.



The system closes the gap between raw port data and confident operational decisions.

System Architecture

Frontend Dashboard

Real-time risk windows, berth views, and routing guidance

Backend API

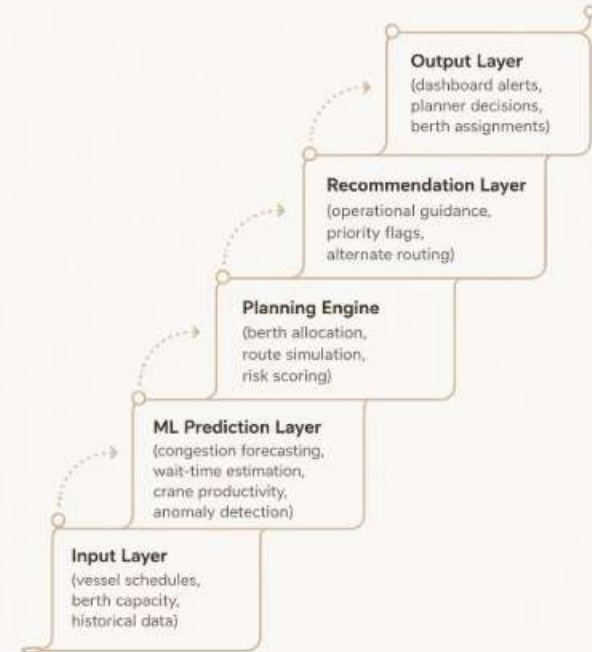
FastAPI layer
orchestrating data flow and business logic

ML Prediction Layer

Congestion risk, wait-time, and crane productivity models

Planning Engine

Berth allocation, simulation, and anomaly detection



Live Congestion Forecasting — In One View

The planner dashboard surfaces the highest-priority operational signals: risk windows, berth availability, and recommended vessel handling sequences — all in a single, coherent interface.



Congestion Risk Window

ML-driven forecast highlights 4–12 hour congestion windows before they materialize



Berth Assignment Support

Ranked berth recommendations based on vessel ETA, draft, and current terminal load



Alternate Routing Guidance

When primary berths are constrained, the system suggests viable alternate handling options

AI & Technology Stack

PortPulse is built on a practical, production-oriented stack — not a prototype. The ML layer is a decision-support capability embedded directly into operational workflows, not a decorative add-on.



Backend

Python · FastAPI ·
SQLAlchemy · PyMySQL ·
Pydantic

ML Layer

scikit-learn · joblib ·
regression & classification
models

Frontend

JavaScript · HTML · CSS ·
API-driven dashboard

Operational Impact

PortPulse shifts port operations from reactive crisis management to proactive, data-informed planning — reducing uncertainty at every decision point.

Before PortPulse

Delayed decisions, unplanned queues, idle berths, and reactive firefighting across vessel operations

With PortPulse

Early congestion signals, smarter berth assignments, coordinated vessel handling, and confident planning

- ❑ **For supervisors and managers:** Faster decisions, improved planning confidence, and stronger coordination across all vessel operations — with a single source of operational truth.



Team Surfers

Hetvi Tank

Team Lead — Architecture oversight and
Documentation

Trusha Patel

Frontend — Dashboard interface, visualization, and
user experience

Jigar Sakhia

Backend & ML — Congestion prediction models, and
ML pipeline

Priyansh Sukhdia

Data — Database Handling